SQL
INTERVIEWER-STYLE PRACTICE

Fresher Data Analyst | English answers + Hindi explanations + practical examples + follow-up cross questions

20 interview scenarios | Practice answering aloud | Use only project details you have actually verified

How to practice

Read the question first and answer aloud before reading the model answer. Learn the concept, not just the wording. For syntax, mention that exact functions may vary by tool or database version. For project questions, describe only work you personally completed.

Interviewer Q1: You have two tables: Employees and Departments. How will you show every employee with their department name?

Suggested English answer: I would use a LEFT JOIN if I need to retain every employee, including those without a matching department. If only matched employees are required, I would use INNER JOIN.

Hindi me samjho: Har employee ko rakhna hai, chahe department match na ho, toh LEFT JOIN. Sirf matching records chahiye toh INNER JOIN.

```
Practical example:
SELECT e.employee_name, d.department_name
FROM Employees e
LEFT JOIN Departments d ON e.department_id=d.department_id;
```

Interviewer follow-up: Why not INNER JOIN?

How to answer: INNER JOIN unmatched employees ko hata dega. Requirement ke hisab se join choose karna chahiye.

Interviewer Q2: How would you find the second-highest distinct salary?

Suggested English answer: I would rank distinct salary values using DENSE_RANK and select rank 2. This handles ties correctly.

Hindi me samjho: Distinct second salary ke liye DENSE_RANK use karunga. Same salary wale employees ko same rank milega.

```
Practical example:
SELECT employee_name, salary
FROM (SELECT employee_name, salary,
DENSE_RANK() OVER(ORDER BY salary DESC) r
FROM Employees) x
WHERE r=2;
```

Interviewer follow-up: What if the interviewer asks for the second row, not second distinct salary?

How to answer: Then ROW_NUMBER with a deterministic tie-breaker may be appropriate; clarify the requirement first.

Interviewer Q3: Find employees whose salary is above their department average.

Suggested English answer: I would calculate the department average using a window function and compare each employee's salary against it.

Hindi me samjho: Har department ka average nikaal kar employee salary se compare karenge.

```
Practical example:
SELECT employee_name, department_id, salary
FROM (SELECT e.*, AVG(salary) OVER(PARTITION BY department_id) dept_avg
FROM Employees e) x
WHERE salary>dept_avg;
```

Interviewer follow-up: Why use PARTITION BY?

How to answer: It calculates a separate average for each department without collapsing employee rows.

Interviewer Q4: How do you identify duplicate email addresses?

Suggested English answer: I would group by email and use HAVING COUNT(*) greater than one. I would inspect the records before deleting anything.

Hindi me samjho: Email group karke count 1 se zyada ho toh duplicate candidate hai. Delete se pehle records verify karunga.

```
Practical example:
SELECT email, COUNT(*) AS cnt
FROM Customers
GROUP BY email
HAVING COUNT(*)>1;
```

Interviewer follow-up: Would you delete all but one automatically?

How to answer: No. First define which record to retain, such as latest verified record, and preserve an audit/backup.

Interviewer Q5: A query is slow. What steps would you take?

Suggested English answer: I would inspect the execution plan, check join keys and indexes, filter early where appropriate, avoid SELECT *, verify data types, and measure changes.

Hindi me samjho: Execution plan, indexes, join keys, unnecessary columns aur filters check karunga; changes ko measure karunga.

Interviewer follow-up: What is an execution plan?

How to answer: It shows how the database intends to execute a query, including scans, joins, and estimated costs.

Interviewer Q6: Explain WHERE and HAVING with an example.

Suggested English answer: WHERE filters source rows before grouping; HAVING filters aggregated groups after GROUP BY.

Hindi me samjho: WHERE grouping se pehle rows filter karta hai; HAVING group banne ke baad aggregate result filter karta hai.

```
Practical example:
SELECT department_id, AVG(salary) avg_salary
FROM Employees
WHERE active=1
GROUP BY department_id
HAVING AVG(salary)>4000;
```

Interviewer follow-up: Can WHERE use AVG(salary)?

How to answer: Normally aggregate conditions belong in HAVING or a subquery/CTE.

Interviewer Q7: How would you calculate monthly revenue?

Suggested English answer: I would group orders by year and month, sum the revenue column, and validate the total against the source.

Hindi me samjho: Order date ke year-month par group karke revenue ka SUM karunga; total source se validate karunga.

```
Practical example:
SELECT EXTRACT(YEAR FROM order_date) yr,
EXTRACT(MONTH FROM order_date) mn,
SUM(order_amount) revenue
FROM Orders
GROUP BY EXTRACT(YEAR FROM order_date), EXTRACT(MONTH FROM order_date);
```

Interviewer follow-up: Does this syntax work in every database?

How to answer: Date extraction syntax varies. SQL Server commonly uses YEAR() and MONTH(); adapt to the database.

Interviewer Q8: What is the difference between UNION and UNION ALL?

Suggested English answer: UNION removes duplicate result rows; UNION ALL preserves them and can avoid the deduplication work.

Hindi me samjho: UNION duplicates hataata hai; UNION ALL duplicates rakhta hai.

Interviewer follow-up: Which would you use for appending monthly sales files?

How to answer: If duplicate removal is not required and schemas align, UNION ALL is commonly appropriate.

Interviewer Q9: Explain a CTE and why you use it.

Suggested English answer: A CTE is a named temporary result defined with WITH. It makes multi-step queries easier to read and maintain.

Hindi me samjho: WITH ke through named result banta hai; complex query ko readable steps me tod sakte hain.

```
Practical example:
WITH dept_sales AS (
SELECT department_id, SUM(sales) total_sales
FROM Orders GROUP BY department_id
)
SELECT * FROM dept_sales WHERE total_sales>10000;
```

Interviewer follow-up: Is a CTE always faster than a subquery?

How to answer: No. It is primarily a readability tool; optimization depends on the database engine and query.

Interviewer Q10: How would you find customers who never placed an order?

Suggested English answer: I can use LEFT JOIN and filter NULL on the order key, or use NOT EXISTS.

Hindi me samjho: Customers ko Orders se LEFT JOIN karke unmatched order key NULL check karunga; NOT EXISTS bhi option hai.

```
Practical example:
SELECT c.customer_id, c.customer_name
FROM Customers c
WHERE NOT EXISTS (SELECT 1 FROM Orders o WHERE o.customer_id=c.customer_id);
```

Interviewer follow-up: Why might NOT EXISTS be useful?

How to answer: It expresses an anti-match condition clearly and avoids duplicate customer rows from multiple orders.

Interviewer Q11: What is a window function?

Suggested English answer: It calculates across a related set of rows while retaining each row, useful for ranking, running totals, and comparisons.

Hindi me samjho: Rows ko collapse kiye bina group-related calculation karta hai.

```
Practical example:
SELECT order_date, sales,
SUM(sales) OVER(ORDER BY order_date) running_sales
FROM DailySales;
```

Interviewer follow-up: How is it different from GROUP BY?

How to answer: GROUP BY typically returns one row per group; window functions preserve detail rows.

Interviewer Q12: How do you handle NULL values?

Suggested English answer: I check whether NULL means unknown, missing, or not applicable, then use IS NULL, COALESCE, or a justified cleaning rule.

Hindi me samjho: NULL ka meaning samajhkar IS NULL/COALESCE ya justified cleaning rule use karunga.

```
Practical example:
SELECT COALESCE(phone,'Not provided') FROM Customers;
```

Interviewer follow-up: Why not use = NULL?

How to answer: NULL represents unknown; SQL uses IS NULL for null checks.

Interviewer Q13: What is a primary key versus a foreign key?

Suggested English answer: A primary key uniquely identifies a row. A foreign key references a key in another table to maintain relationships.

Hindi me samjho: Primary key row ki unique identity; foreign key doosri table ki key ko reference karti hai.

Interviewer follow-up: Can a foreign key contain NULL?

How to answer: It can, if the column permits NULL and the relationship is optional.

Interviewer Q14: Explain ACID properties.

Suggested English answer: Atomicity, Consistency, Isolation, and Durability describe transaction reliability: all-or-nothing, valid states, controlled concurrency, and persistence.

Hindi me samjho: Transaction ya toh complete ya rollback; rules valid rahen, concurrent work controlled ho, commit data persist kare.

Interviewer follow-up: Give a banking example.

How to answer: A transfer should debit one account and credit another together, not leave only one side completed.

Interviewer Q15: What is normalization?

Suggested English answer: Normalization structures tables to reduce unnecessary duplication and update anomalies while preserving relationships.

Hindi me samjho: Data duplication aur insert/update/delete anomalies kam karne ke liye tables ko organize karna.

Interviewer follow-up: Can too much normalization affect reporting?

How to answer: Reporting may need more joins; dimensional models can denormalize selected attributes for analytical usability.

Interviewer Q16: How do you safely update records?

Suggested English answer: I first preview the exact rows with SELECT using the same WHERE condition, use a transaction when supported, update, and verify affected rows.

Hindi me samjho: UPDATE se pehle same condition ka SELECT; transaction ho toh use; baad me affected rows verify.

```
Practical example:
UPDATE Employees SET salary=salary*1.05 WHERE department_id=10;
```

Interviewer follow-up: What is the danger of missing WHERE?

How to answer: It may update every row, so preview and transaction safeguards matter.

Interviewer Q17: What does CASE WHEN do?

Suggested English answer: It creates conditional output, similar to if/else logic.

Hindi me samjho: Condition ke basis par category ya flag banata hai.

```
Practical example:
SELECT employee_name,
CASE WHEN salary>=6000 THEN 'High' ELSE 'Other' END salary_band
FROM Employees;
```

Interviewer follow-up: Can CASE be used inside SUM?

How to answer: Yes, for conditional aggregation, such as SUM(CASE WHEN status='Paid' THEN amount ELSE 0 END).

Interviewer Q18: How do you validate a SQL result before sharing it?

Suggested English answer: I check row counts, nulls, duplicates, join coverage, totals against source data, and sample records; I document assumptions.

Hindi me samjho: Row count, NULL, duplicates, join coverage aur totals validate karunga; assumptions note karunga.

Interviewer follow-up: What if totals differ from Excel?

How to answer: Check filters, date boundaries, duplicate joins, data refresh timing, and definition differences.

Interviewer Q19: What is the difference between DELETE, TRUNCATE, and DROP?

Suggested English answer: DELETE removes rows and can filter; TRUNCATE clears table rows; DROP removes the table object. Transaction and identity behavior vary by DBMS.

Hindi me samjho: DELETE selected rows; TRUNCATE data clear; DROP table object remove. DBMS behavior vary kar sakta hai.

Interviewer follow-up: Can TRUNCATE be rolled back?

How to answer: That depends on the database engine and transaction context; don't assume identical behavior everywhere.

Interviewer Q20: How would you explain your SQL skill to an interviewer?

Suggested English answer: I can extract relevant data, join tables, filter and aggregate metrics, use subqueries and window functions, and validate outputs for reporting.

Hindi me samjho: Main data extract, joins, filters, KPI aggregation aur window functions use karke report se pehle output validate kar sakta hoon.

Interviewer follow-up: What would you do if you don't know a function?

How to answer: Clarify the requirement, consult trusted documentation, test on sample data, and validate the result.

Final interview reminder: Pause, clarify the requirement, explain your approach, and validate the result. It is acceptable to say you would verify a detail rather than guess.